

Avatar Diversity in Social VR: From the Perspectives of Compatible Outfit Concentration and Customization Range

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ABSTRACT

In Social VR, avatars and outfits are 3D models through which users present embodied representations and update self-expression across changing contexts and relationships. The practical viability of an avatar is therefore shaped not only by the availability of avatar types but also by how easily users can continue updating their appearance after selection through outfits and related resources. In UGC-oriented ecosystems such as VRChat, many transactions for avatars and outfits occur via external marketplaces, and outfit listings commonly specify which avatar(s) they are compatible with because compatibility is not universal across rigs and settings. This market structure makes the number of outfit listings explicitly marked as compatible with an avatar an observable proxy for post-selection customization resources. In this preliminary analysis, we focus on BOOTH, a major marketplace in the VRChat ecosystem, and analyze a single-time-point snapshot dataset as of December 6, 2025, restricting measurement to avatars with at least five compatible outfit listings. We characterize the distribution of compatible outfit counts at two levels: the avatar level and the creator level after normalizing creator names and aggregating counts by creator. The results reveal a pronounced long-tail structure and a two-stage pattern of concentration: compatible outfit counts cluster around a limited set of avatars with many compatible listings, and the skew becomes more pronounced when aggregated at the creator level, indicating that a relatively small number of creators account for a large share of compatible outfits. Because the data capture one point in time, we do not infer causal direction; instead, we present mechanism hypotheses consistent with the observed patterns. On the supply side of avatar-compatible outfit creation, avatar-specific compatibility work can create incentives to concentrate effort on avatars that are more likely to be widely used. On the demand side, as users build a larger collection of outfits tied to a given avatar, switching to avatars with fewer compatible listings can become increasingly costly, and users may select avatars while anticipating future outfit availability. Together, these dynamics may place users in a tradeoff between avatars with many compatible outfits and those with fewer, potentially reinforcing convergence on a smaller set of commonly adopted avatars and narrowing morphological diversity in actual use.

Index terms: VRChat, Avatar, UGC, Customization

1 INTRODUCTION

Social VR platforms exemplified by VRChat provide an environment in which users interact in ways close to face-to-face, premised on immersive shared spaces and embodied presence [1].

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In such environments, avatars are not merely superficial appearances but function as bodies perceived by others, strongly shaping self-expression, belonging, sense of safety, and positioning within communities. Empirical studies on self-presentation in Social VR have shown that users present themselves through avatars and that such presentations influence interaction and relationship formation.

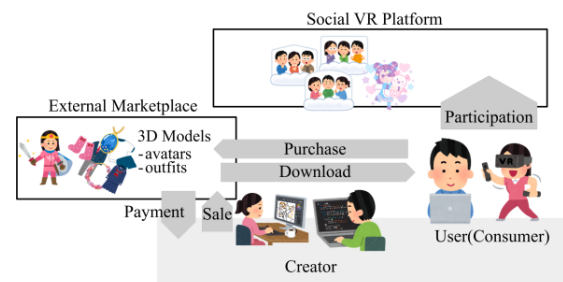


Fig 1. Flow of Avatar Usage in the Social VR

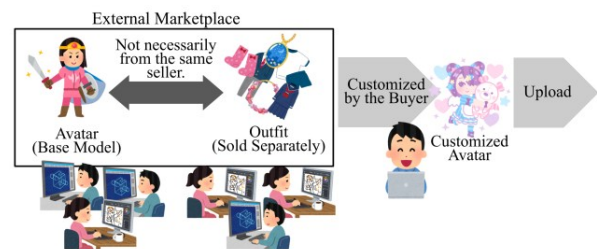


Fig 2. Avatar and Outfit Relationship

However, discussions of “avatar diversity” often collapse into whether there are many avatar types available, overlooking practical usability. For diverse self-expression to be realized in Social VR, it is insufficient that diverse avatars merely exist. After a user who lacks the technical skills to create 3D models selects an avatar, the key question is whether they can modify their expression in accordance with contexts and relationships. That is, whether they can access “parts that extend expression”, such as outfits, accessories, and hairstyles (hereafter, collectively “outfits”).

In user-generated content (UGC)-oriented Social VR, users often rely on marketplaces to obtain expressive digital goods. In the VRChat ecosystem, many transactions for avatars and outfits are conducted via external marketplaces (e.g., BOOTH) (Fig. 1,2). While this division of labor supports the growth of the UGC ecosystem, it also introduces compatibility constraints and associated costs. Here, “UGC-oriented” refers to Social VR platforms where user-created content is central to the experience, including the creation and distribution of worlds and expressive assets. VRChat exemplifies this orientation, as user-created worlds and avatar-related assets are core to its ecosystem.

Moreover, avatars and outfits are often sold separately by different entities (Fig. 2). While this division of labor supports the

growth of the UGC ecosystem, it also introduces compatibility constraints and associated costs.

In VRChat, for example, because skeletal rigs and settings differ by avatar, outfits are not universally compatible even within the same platform. They must be adjusted to fit specific avatars and are typically sold in the form “3D model of a school-uniform outfit: compatible with avatar XX” [2].

These compatibility costs tend to direct outfit creators’ investment toward popular avatars, and on the user side, switching costs that entail losing existing outfits discourage migration. Concretely, when a user who owns many outfits for Avatar A moves to Avatar B, they incur the burden of having to repurchase outfits. Under this structure, the practical effectiveness of avatar diversity cannot be captured by the “number of avatars” alone. Rather, what matters is how many and how diverse outfits and modification resources a given avatar receives, in other words, how customization range is distributed across avatars. This paper focuses on this point, uses the “number of outfits compatible with an avatar” observed in external markets as a proxy for an avatar’s customization range, and quantitatively describes the distribution of customization range among top avatars and creators.

The contributions of this paper are threefold:

(i) It proposes a framework for observing diversity in Social VR not as the number of existing avatars, but as customization range (the range of compatible outfits, operationalized as the count of compatible outfits).

(ii) Using publicly available information from external markets related to VRChat, it identifies avatars with the highest numbers of compatible outfits and presents the distribution aggregated at the avatar-creator level.

(iii) It summarizes key features of the distribution and discusses implications for diversity in avatars and outfits.

2 AVATAR DIVERSITY IN SOCIAL VR

2.1 Self-Expression and Avatars

Avatars are a medium of self-expression and also serve as bodies that underpin social interaction [3,4]. In Social VR, it has been reported that, through avatars, users present not only selves close to their offline identities but also idealized and exploratory (experimental) self-images depending on context. These modes of self-presentation, mediated through others’ perceptions and ongoing interactions, can be associated with how relationships are formed and how users engage in community activities [1,5,6,7].

In this context, “diversity” should not be reduced to the mere breadth of available options. It is also important whether users with particular attributes or experiences can participate safely with a bodily representation that feels authentic to them, and whether they can sustain that representation and update it as situations change [8].

2.2 Relationship Between Avatars and Outfits

A defining feature of UGC-oriented Social VR is that expressive presentation is realized through the coupling of the avatar itself with complementary goods such as outfits. A wider range of outfits and accessories can increase an avatar’s attractiveness, encourage user participation, and potentially generate a positive feedback loop: as more users participate, outfit creators can better observe demand, which in turn leads to the creation of more outfits. In addition, prior research suggests that avatar attractiveness can positively affect impression formation [9], and that because users tend to design and create avatars based on their own personality, a wider range of avatar customization is important for enhancing a platform’s competitiveness [10,11].

Critically, this feedback loop is not neutral; compatibility constraints can introduce systematic bias. In VRChat, avatars and outfits are traded on external e-commerce sites (such as BOOTH), where search, payment, and downloading are completed within the external site’s interface (although VRChat’s in-platform “Avatar Marketplace” allows the purchase of avatars alone, albeit without secondary customization [12]). Moreover, outfits are not universally compatible: because skeletal rigs and configuration settings differ across avatars, fitting an outfit to a given avatar typically requires avatar-specific adjustment, and compatibility is therefore often stated explicitly in product descriptions.

Therefore, to evaluate the practical diversity of avatar-based expression in Social VR, it is insufficient to discuss only the diversity of avatars in isolation (for example, the gender or race of avatars on sale). It is also meaningful to consider indicators such as where opportunities for secondary customization via outfits are concentrated, as observed through the distribution of outfit supply in external markets, and in particular, which avatars and which avatar creators attract these customization opportunities.

3 RESEARCH METHODS

3.1 Basic Design of Data Collection

This study collects and analyzes data on the relationship between avatars and outfits on an external marketplace, focusing on a single marketplace as the unit of observation. The objects of analysis are “3D models used in VRChat (avatars and outfits).” As the sales platform, we selected BOOTH [13], which is operated by pixiv Inc. in Japan.

BOOTH is a platform for creators, and in 2024 its transaction value for the “3D model” category was JPY 5.8 billion [14]. The dataset used in this study reflects the status as of December 6, 2025.

3.2 Units of Observation

(A) Avatar level: For each avatar, we obtain the number of compatible outfits. Compatible outfits refer to outfits for which compatibility with the focal avatar is explicitly stated on the product page or related materials. Given that outfit compatibility is not universal, the marketplace structure, in which outfits are sold as “compatible with avatar XX,” is empirically observable. The number of compatible outfits was provided by the avatar and outfit information site “Avatar Network” [15]. To focus on avatars that plausibly offer at least a minimum set of customization options, we restrict measurement to avatars with five or more compatible outfits. Note that a single outfit may be compatible with multiple avatars.

(B) Creator level: From a diversity perspective, it is important to consider not only the prominence of particular avatars, but also the possibility that specific avatar creators hold multiple popular avatars and thereby attract a disproportionate accumulation of outfit supply. To examine portfolio composition, we additionally decompose each creator’s total compatible-outfit count into the contributions of the creator’s top five avatars (ranked by compatible-outfit counts within the creator) and the remaining avatars aggregated as “Other” (used for visualization in Fig. 3).

We compute each avatar’s (or creator’s) share as the number of compatible outfits attributed to it divided by the total across all included avatars (or creators). HHI is calculated as the sum of squared shares (reported on a 0–10,000 scale), capturing top-heavy concentration, while the Gini coefficient summarizes overall unevenness in the distribution.

Limitation: The number of compatible outfits is a proxy for the quantity of customization range; it does not directly measure diversity across representational categories (e.g., disability representation, gender representation, racial representation).

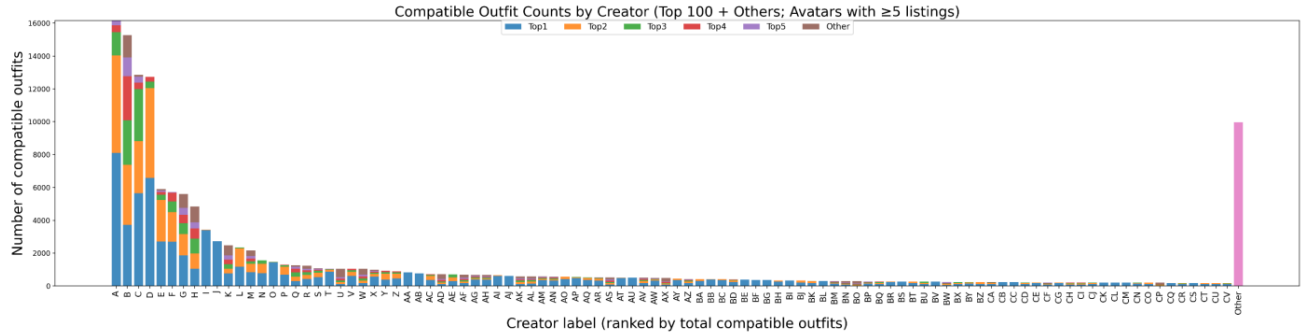


Fig 3. Compatible Outfit Counts by Creator

4 RESULTS

In this section, we summarize the “number of compatible outfits” observed in the external market at two levels: (i) the pre-disambiguation avatar level and (ii) the post-disambiguation creator level (Table 1). Because the dataset is restricted to avatars with five or more compatible outfits, all statistics reported below pertain to this conditional population.

Table 1. Descriptive Statistics of Compatible Outfit Counts, by Avatar and by Creator

Metric	Avatar level	Creator level
Observations	961	382
Total compatible outfits	144,889	144,889
Max	8,097	16,155
Median	26	39
Mean	150.77	379.29
HHI	164.74	481.14
Gini	0.819	0.862

Supplementary quantiles indicate that, at the avatar level, the 95th percentile is 501 and the 99th percentile is approximately 2,897. At the creator level, the 95th percentile is approximately 1,056 and the 99th percentile is approximately 7,204, implying a heavier upper tail.

A rank-ordered visualization of compatible outfits by creator illustrates two layers of concentration (Fig. 3). Creators are ranked by their total number of compatible outfits, and for each of the top creators the total is decomposed into the contributions of the creator’s top five avatars (Top1–Top5, ranked within the creator) and the remainder (Other). Because creators are ranked from highest to lowest, the largest values appear at the leftmost end, and the long tail extends to the right. For readability, only the top 100 creators are shown individually; all remaining creators are aggregated into a single “Other” bar. This visualization complements summary indices (Gini and HHI) by making the long-tail structure across creators and the within-creator dependence on a small set of avatars jointly observable.

4.1 Distribution at the avatar level

At the avatar level, compatible outfits are concentrated in a small number of avatars. Examples of top values include 8,097, 6,584, 5,930, 5,647, 5,456, 3,724, 3,652, 3,390, 3,169, and 3,169, forming the upper tail of the distribution.

By thresholds, among 961 avatars, 49 have 500 or more compatible outfits, 25 have 1,000 or more, 16 have 2,000 or more, and 5 have 5,000 or more, indicating that a small set of top avatars effectively drive the “compatibility ecosystem” within the market.

4.2 Distribution and concentration at the creator level

When aggregated at the creator level, the skew becomes even more pronounced. The numbers of compatible outfits for the top creators are: A: 16,155; B: 15,270; C: 12,846; D: 12,733; E: 5,907; F: 5,719; G: 5,595; H: 4,827; I: 3,421; and J: 2,716 (Fig. 3).

To provide an intuitive sense of concentration, we compute cumulative shares of the total number of compatible outfits. The top 1 creator accounts for 11.15%, the top 2 for 21.69%, the top 4 for 39.34%, the top 10 for 58.80%, the top 50 for 83.93%, and the top 100 for 93.13%, showing strong accumulation among the upper tier.

Beyond between-creator concentration, Fig. 3 also highlights heterogeneity in portfolio structure within top creators. For some top creators, a substantial portion of their totals is accounted for by their Top1–Top2 avatars, whereas others exhibit a larger “Other” segment, suggesting a more diversified portfolio of moderately supported avatars.

5 DISCUSSION

5.1 Diversity as Customization Range

If diversity is defined as “the number of avatar types,” the discussion tends to converge on whether particular representations (e.g., gender or race) are present. In contrast, in UGC-oriented Social VR, the self-expression that users engage in on a daily basis is realized through iterative updates using outfits and other customization resources. Therefore, the practical effectiveness of avatar diversity should also be evaluated in terms of how many and what kinds of outfits and customization resources users can access after choosing a particular avatar, that is, an avatar’s customization range. In this paper, we focus on this usability dimension of diversity (the allocation of updateability), rather than on representational categories per se, and treat compatible-outfit availability as a practical proxy for customization range that supports sustained self-presentation.

Based on our data, customization range is unevenly distributed across avatars. Even at the avatar level, the Gini coefficient is 0.819, indicating that a small number of top avatars occupy the upper tail of outfit supply. When creators are disambiguated and the data are aggregated at the creator level, the Gini coefficient increases to 0.862 and the HHI rises from 164.74 to 481.14, suggesting that the skew becomes even stronger. This implies that not only concentration on a single popular avatar, but also the accumulation of outfit supply around creators who hold multiple popular avatars, substantively shapes the allocation of customization range. On the supply side, outfit creators may infer that new avatars released by creators with popular portfolios will sell well, making outfit production for those avatars attractive because it targets a large potential customer base. On the demand side, users may also infer

that new avatars released by such creators will likely attract abundant compatible outfits.

5.2 Mechanisms Underlying Concentration

Our results indicate that the concentration of compatible outfits emerges in two stages: at the avatar level and at the creator level. While concentration is substantial at the avatar level, it is also evident at the creator level. Accordingly, analysis should go beyond the statement “outfits accumulate around popular avatars” and also consider that “outfits accumulate around creators (and their portfolios) that produce popular avatars,” examining both supply-side and demand-side dynamics. Because the present analysis is based on a cross-sectional snapshot of compatibility counts, we do not identify causal direction or disentangle supply-side targeting from demand-side selection. Accordingly, we frame the following as mechanism hypotheses that are consistent with the observed skew and derive testable implications for future longitudinal or behavioral analyses.

(1) Supply-side mechanism: compatibility costs and complementor targeting.

Because outfits require avatar-specific compatibility work (e.g., adjustments to rigs and weights), adding support for additional avatars entails nontrivial costs for outfit creators. As a result, creators may be more likely to prioritize targets with higher expected returns on investment. In our data, at the creator level, the top 10 creators account for 58.80% of the total number of compatible outfits, the top 50 account for 83.93%, and the top 100 account for 93.13%, indicating that support is heavily concentrated in a small set of creators. At the avatar level, only 49 avatars have 500 or more compatible outfits, and only 5 avatars have 5,000 or more, suggesting that avatars with a very broad compatible-outfit range are limited. These highly compatible avatars may become salient targets for outfit creators because they offer a larger prospective customer base. If such targeting occurs, it could further amplify concentration in supply, which is consistent with the observed long-tail structure and may imply comparatively narrower customization range for peripheral avatars.

A further supply-side mechanism is portfolio spillover: when a creator already holds multiple supported avatars, outfit creators can expect an installed base and reduced uncertainty for new releases by the same creator. This mechanism would predict faster early accumulation of compatible outfits for newly released avatars by portfolio-holding creators compared with avatars from peripheral creators. While we do not test this timing claim in the present snapshot, a testable implication is that time-to-first compatibility and early growth rates would be expected to be shorter for avatars released by creators in the upper tail.

(2) Demand-side mechanism: outfit asset ties and switching costs.

On the demand side, the more outfits accumulate as assets tied to a specific avatar, the more changing avatars entails losses through the depreciation of the existing outfit stock [16]. Experimental evidence further suggests that avatar customization can increase purchase intention for virtual fashion items via avatar embodiment and subsequent psychological ownership [17]. This implies that the cost of switching may be not only monetary (repurchasing outfits) but also psychological when users perceive avatar-linked items as “theirs.” In addition, users may choose avatars while anticipating future customization support, namely, whether continued outfit supply can be expected [16]. Once a set of highly compatible avatars emerges, users may increasingly prefer those avatars because of the expectation of ongoing outfit availability [18]. Consequently, even if diversity in the sense of the sheer number of existing avatars is maintained, the capacity for repeated expression updates via customization may become biased

toward highly compatible avatars, and the diversity sustained after selection may shrink.

5.3 Implications for Diversity

When customization range is unevenly distributed across avatars, the diversity observed within Social VR is also likely to be skewed. As users repeatedly update their expression through outfits, they accumulate outfit assets and find it progressively harder to switch avatars. As a result, a small set of avatars with a broad compatible-outfit range may be preferred as de facto standard bodies, while peripheral avatars may exist but become less likely to be chosen.

Users may face a choice between “a popular avatar with a wide range of customization” and “an unpopular avatar with a narrow range of customization.” Because customization is an essential element for sustaining self-expression and developing attachment to the platform, users have strong incentives to choose the former as a rational decision, which may in turn reinforce the concentration of body representations.

In the short run, this skew can be understood as individually rational behavior: users choose avatars that maximize expected customization range, and outfit creators allocate effort where compatibility work is most likely to be recouped. In the medium run, however, the same logic can raise barriers for peripheral avatars, because limited early support makes them less attractive to both creators and users, reinforcing a self-fulfilling disadvantage. Even if the market offers many avatar options, uneven customization opportunities may lead to a smaller set of avatars being widely adopted in practice. If customization resources concentrate around a limited set of base bodies, users may face incentives to select those bodies to sustain ongoing self-presentation. As a result, the diversity of underlying morphologies in actual use may decrease, including variation in body build and in non-human forms such as animal-like or robotic bodies.

6 CONCLUSION

This paper conceptualized avatar diversity in UGC-oriented Social VR not as the sheer number of available avatars, but as customization range that enables users to update their self-presentation after selecting an avatar. We operationalized customization range using compatible-outfit listings from an external marketplace. In such external markets, avatars and outfits circulate through a division of labor, and because outfits require avatar-specific compatibility work (e.g., rig adjustments), supply may be drawn toward targets with higher expected prospects for recouping investment.

In analyses restricted to avatars with five or more compatible outfits, customization range exhibits a pronounced long-tail distribution, and the skew is amplified when records are disambiguated and aggregated at the creator level. The cumulative share of total compatible outfits reaches 58.80% for the top 10 creators and 93.13% for the top 100, suggesting that a small number of creators substantively shape the distribution of customization range. These findings suggest that if diversity is assessed only in terms of “what exists,” researchers may overlook systematic bias in the customization support that is actually sustained after avatar selection.

This study does not seek to characterize the observed concentration of compatible-outfit availability across specific avatars and creators as inherently negative at the present stage. From a user-utility perspective, the existence of avatars with exceptionally broad customization options can be beneficial, as it expands feasible expression and reduces search and coordination costs for both users and complementary creators. Moreover, based on a single cross-sectional snapshot, we do not provide evidence that the current distribution reflects anticompetitive conduct, nor do we identify clear, concrete user harms attributable to uneven

customization availability at this time. Normative assessment therefore remains an empirical question that should be examined in future work using longitudinal market data and, where possible, behavioral evidence on user choice, switching costs, and creator targeting. Nonetheless, the descriptive fact that meaningful concentration exists is important in its own right, and the hypothesized mechanisms discussed in this paper imply that the distribution may further tip toward a smaller set of avatars and creator portfolios over time. Establishing whether such dynamics ultimately enhance or undermine practical diversity, and under what conditions, is a key research agenda for UGC-oriented Social VR ecosystems.

Future work should measure not only the magnitude of customization range but also its allocation across representational categories, directly testing which forms of representation are more readily extensible and which are less so. In addition, evaluating the extent to which interventions such as compatibility support, information standardization, discoverability design, and support measures for peripheral representations can mitigate the skew in customization range will be important for implementing inclusive diversity in Social VR.

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